

30 CHAPTERWISE IMPORTANT QUESTIONS

Science (086) | CBSE 2026-27

The Human Eye and the Colourful World

CBSE Science 2026-27 | Board-examinable

30 original revision questions with answers: short-answer, competency, assertion-reason and long-answer practice. Set is based on the current syllabus and official question-paper style; it is not a prediction or an official CBSE paper.

Question 1

What causes myopia, and which lens corrects it?

Answer:

In myopia, distant objects focus in front of the retina, often because the eyeball is too long or the eye lens too powerful. A concave lens diverges incoming rays so the image forms on the retina.

Question 2

How does the eye accommodate objects at different distances?

Answer:

Ciliary muscles change the curvature and focal length of the eye lens. The lens adjusts so light from near or distant objects is focused on the retina.

Question 3

Why does a prism disperse white light?

Answer:

Different colours have different wavelengths and refract by different amounts in glass. The colours separate into a spectrum as they emerge from the prism.

Question 4

What is the central idea or skill tested in The Human Eye and the Colourful World?

Answer:

The central focus is: In myopia, distant objects focus in front of the retina, often because the eyeball is too long or the eye lens too powerful. This is the starting point for a strong answer from this chapter.

Question 5

Write two relevant points for a short answer based on The Human Eye and the Colourful World.

Answer:

First point: In myopia, distant objects focus in front of the retina, often because the eyeball is too long or the eye lens too powerful. Second point: Ciliary muscles change the curvature and focal length of the eye lens. Link the two points clearly instead of listing them without explanation.

Question 6

How does the first key idea in The Human Eye and the Colourful World lead to an important result or conclusion?

Answer:

Begin with the key idea: In myopia, distant objects focus in front of the retina, often because the eyeball is too long or the eye lens too powerful. Then connect it to the result: Ciliary muscles change the curvature and focal length of the eye lens.

The Human Eye and the Colourful World - Continued

Question 7

What evidence, calculation, incident or concept best supports the first main idea in The Human Eye and the Colourful World?

Answer:

Use this supporting detail: In myopia, distant objects focus in front of the retina, often because the eyeball is too long or the eye lens too powerful. A concave lens diverges incoming rays so the image forms on the retina. The relevant detail should be stated before giving the final conclusion.

Question 8

Explain the importance of the second major idea in The Human Eye and the Colourful World.

Answer:

Ciliary muscles change the curvature and focal length of the eye lens. The lens adjusts so light from near or distant objects is focused on the retina. Its importance lies in showing how the chapter develops the topic beyond a simple fact or definition.

Question 9

What common mistake should a student avoid while answering a question from The Human Eye and the Colourful World?

Answer:

Do not give only a conclusion. Explain the reasoning with a correct detail, such as: Different colours have different wavelengths and refract by different amounts in glass.

Question 10

Compare the first and second important ideas in The Human Eye and the Colourful World.

Answer:

The first idea is: In myopia, distant objects focus in front of the retina, often because the eyeball is too long or the eye lens too powerful. The second idea is: Ciliary muscles change the curvature and focal length of the eye lens. A complete comparison states both the difference and the connection between them.

Question 11

Show a cause-and-effect link from The Human Eye and the Colourful World.

Answer:

Cause or starting point: In myopia, distant objects focus in front of the retina, often because the eyeball is too long or the eye lens too powerful. Effect or outcome: Ciliary muscles change the curvature and focal length of the eye lens. Use connecting words such as 'therefore', 'as a result', or 'thus'.

Question 12

What is the most useful sequence for explaining the main content of The Human Eye and the Colourful World?

Answer:

A clear sequence is: In myopia, distant objects focus in front of the retina, often because the eyeball is too long or the eye lens too powerful. Next, Ciliary muscles change the curvature and focal length of the eye lens. Finally, Different colours have different wavelengths and refract by different amounts in glass.

The Human Eye and the Colourful World - Continued

Question 13

Give a 40-50 word answer outline for an important question from The Human Eye and the Colourful World.

Answer:

Start with the direct answer: In myopia, distant objects focus in front of the retina, often because the eyeball is too long or the eye lens too powerful. Add one supporting point: Ciliary muscles change the curvature and focal length of the eye lens. End with the outcome or message: Different colours have different wavelengths and refract by different amounts in glass.

Question 14

Which scientific observation, equation, diagram, or principle should be written accurately in The Human Eye and the Colourful World?

Answer:

Use an accurate scientific statement: In myopia, distant objects focus in front of the retina, often because the eyeball is too long or the eye lens too powerful. Then link it to the observation, process, or result asked in the question.

Question 15

How can an examiner see that you understand, rather than memorise, The Human Eye and the Colourful World?

Answer:

Connect the idea to its reason or result. For instance: Ciliary muscles change the curvature and focal length of the eye lens. Then explain why this point matters in the chapter's context.

Question 16

State one inference that can be drawn from the main situation or concept in The Human Eye and the Colourful World.

Answer:

A sound inference is that different colours have different wavelengths and refract by different amounts in glass. This inference must be supported by a relevant point from the chapter.

Question 17

Frame a short competency-based response using the first two ideas in The Human Eye and the Colourful World.

Answer:

Apply the concept by first stating: In myopia, distant objects focus in front of the retina, often because the eyeball is too long or the eye lens too powerful. Then justify the response with: Ciliary muscles change the curvature and focal length of the eye lens.

Question 18

What would make an answer from The Human Eye and the Colourful World incomplete?

Answer:

An answer is incomplete if it omits the explanation or final implication. Include both: In myopia, distant objects focus in front of the retina, often because the eyeball is too long or the eye lens too powerful. Also add: Different colours have different wavelengths and refract by different amounts in glass.

The Human Eye and the Colourful World - Continued

Question 19

Prepare an Assertion-Reason style explanation based on The Human Eye and the Colourful World.

Answer:

Assertion: In myopia, distant objects focus in front of the retina, often because the eyeball is too long or the eye lens too powerful. Reason: Ciliary muscles change the curvature and focal length of the eye lens. Check whether the reason actually explains the assertion before choosing an option.

Question 20

How should you justify an answer from The Human Eye and the Colourful World in two clear steps?

Answer:

Step 1: state the concept or event - In myopia, distant objects focus in front of the retina, often because the eyeball is too long or the eye lens too powerful. Step 2: support it with the explanation or result - Ciliary muscles change the curvature and focal length of the eye lens.

Question 21

What answer structure is suitable for a 3-mark question from The Human Eye and the Colourful World?

Answer:

Use three connected points: In myopia, distant objects focus in front of the retina, often because the eyeball is too long or the eye lens too powerful. Ciliary muscles change the curvature and focal length of the eye lens. Different colours have different wavelengths and refract by different amounts in glass. Keep the points specific and write them in a logical order.

Question 22

What answer structure is suitable for a 5-mark question from The Human Eye and the Colourful World?

Answer:

Give an introduction, develop these key points - In myopia, distant objects focus in front of the retina, often because the eyeball is too long or the eye lens too powerful. Ciliary muscles change the curvature and focal length of the eye lens. - and close with: Different colours have different wavelengths and refract by different amounts in glass. Add a diagram, equation, example, or textual reference where relevant.

Question 23

How can the ideas in The Human Eye and the Colourful World be applied to a new situation or case study?

Answer:

Identify the relevant principle first: In myopia, distant objects focus in front of the retina, often because the eyeball is too long or the eye lens too powerful. Then use the chapter's reasoning to explain the case: Ciliary muscles change the curvature and focal length of the eye lens.

The Human Eye and the Colourful World - Continued

Question 24

Which detail from The Human Eye and the Colourful World is most likely to earn an extra mark when explained correctly?

Answer:

A useful detail is: Different colours have different wavelengths and refract by different amounts in glass. Explain it in context instead of using it as an isolated statement.

Question 25

Write a brief concluding statement for an answer on The Human Eye and the Colourful World.

Answer:

A suitable conclusion is: Different colours have different wavelengths and refract by different amounts in glass. This conclusion should follow naturally from the explanation rather than introduce a new point.

Question 26

Make a three-point revision note for The Human Eye and the Colourful World.

Answer:

1. In myopia, distant objects focus in front of the retina, often because the eyeball is too long or the eye lens too powerful. 2. Ciliary muscles change the curvature and focal length of the eye lens. 3. Different colours have different wavelengths and refract by different amounts in glass. Revise these points with the chapter's examples, terms, and practice questions.

Question 27

How would you self-check an answer from The Human Eye and the Colourful World before submitting it?

Answer:

Check that you have answered the exact question, used the key idea - In myopia, distant objects focus in front of the retina, often because the eyeball is too long or the eye lens too powerful. - and supported it with a reason, example, calculation, or outcome such as: Ciliary muscles change the curvature and focal length of the eye lens.

Question 28

Why does The Human Eye and the Colourful World remain important for revision?

Answer:

It combines factual recall with explanation and application. The main takeaways are: In myopia, distant objects focus in front of the retina, often because the eyeball is too long or the eye lens too powerful. Ciliary muscles change the curvature and focal length of the eye lens. Different colours have different wavelengths and refract by different amounts in glass.

Question 29

Create a final 60-80 word answer plan for a long question from The Human Eye and the Colourful World.

Answer:

Opening: In myopia, distant objects focus in front of the retina, often because the eyeball is too long or the eye lens too powerful. Development: Ciliary muscles change the curvature and focal length of the eye lens. Conclusion: Different colours have different wavelengths and refract by different amounts in glass. Use complete sentences and include all required evidence, steps, or references.

The Human Eye and the Colourful World - Continued

Question 30

What is one last-minute revision question you should ask yourself about The Human Eye and the Colourful World?

Answer:

Ask: 'Can I explain why this is true or important?' A correct self-answer should include: In myopia, distant objects focus in front of the retina, often because the eyeball is too long or the eye lens too powerful. and Ciliary muscles change the curvature and focal length of the eye lens.

Sources: [CBSE Science curriculum 2026-27](#); [CBSE Class X sample papers and marking schemes 2026-27](#); [NCERT textbooks](#).

ONE SHOT MASTERY